

EXPERIMENTAL AND WEAR ANALYSIS OF SS-304 ACTIVATED-TIG WELDED JOINTS

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in

PRODUCTION AND INDUSTRIAL ENGINEERING



NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

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CERTIFICATE

This is to certify that the thesis entitled, “**EXPERIMENTAL AND WEAR ANALYSIS OF SS-304 ACTIVATED-TIG WELDED JOINTS**” is being submitted by

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to the National Institute of Technology Kurukshetra, for partial fulfillment of B.Tech (Production and Industrial Engineering) is a record of the bonafide project work carried out by them. They have worked under my supervision for the submission of this technical report, which to our knowledge has reached the requisite standard. This is further certified that the submitted work reached the standard to award the B.Tech Degree in Production and Industrial Engineering. The thesis or any part thereof has not been presented or submitted to any other University or Institute for any degree or diploma.

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We, **Ankush** (123107006), **Rinshu kant**(123107056) & **Axit Meena** (123107019)students of B.Tech(Mechanical Engineering), hereby declare that the Project Dissertation titled- **“EXPERIMENTAL AND WEAR ANALYSIS OF SS-304 ACTIVATED-TIG WELDED JOINTS”** which is submitted by us to the Department of Mechanical Engineering, NIT Kurukshetra in fulfillment of the requirement for awarding of the Bachelor of Technology degree, is not copied from any source without proper citation.This work has not previously formed the basis for the award of any Degree, Diploma, Fellowship, or other similar title or recognition.

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ABSTRACT

This project investigates the application of Activated Tungsten Inert Gas (A-TIG) welding for achieving single-pass, through-thickness welds on 8 mm thick AISI 304 austenitic stainless steel plates—an alloy extensively used in aerospace, automotive, and chemical industries due to its superior corrosion resistance, mechanical strength, and high-temperature tolerance. Traditional Gas Tungsten Arc Welding (GTAW) processes, while providing high-quality welds, suffer from limitations in penetration depth and are inefficient for thick sections, necessitating edge preparation and multiple passes. A-TIG welding, enhanced by oxide-based activating fluxes such as SiO_2 , TiO_2 , and CrO_3 , enables deeper penetration by altering weld pool dynamics through surface tension gradients and arc constriction mechanisms driven by Marangoni convection and Lorentz forces. The study evaluates the performance of individual activated fluxes at welding currents of 150 A and 170 A while maintaining a constant welding speed of 60 mm/min. Preliminary investigations reveal that CrO_3 , due to its strong oxidizing nature and ferrite-stabilizing effect, achieves full 8 mm penetration at both current levels. TiO_2 and SiO_2 also show significant depth improvements, with maximum penetration observed at 170 A. Microstructural analyses indicate phase transformations, dendritic formations, and ferrite-austenite interactions influenced by the flux composition and thermal gradients. The findings underscore the effectiveness of A-TIG in improving productivity and weld quality

without edge preparation, especially using CrO_3 flux. This research provides a foundation for optimizing flux selection and process parameters in aerospace-grade stainless steel applications, contributing to cost-effective, high-performance welding solutions in critical structural components.

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List of Acronyms

MIG -Metal Inert Gas
A-TIG- Activated Tungsten Inert Gas
GTAW- Gas Tungsten Arc Welding
AISI- American Iron and Steel Institute
FA- Ferrite Austenite
ASTM- American Society for Testing and Materials
AutoCAD- Automatic Computer-Aided Design
CNC- Computer Numerical Control
EDM- Electrical Discharge Machining
UTM- Universal Testing Machine
UTS- Ultimate Tensile Strength
DOP- Depth of Penetration

Chapter-1

Introduction

- **Origin for the Project**

Stainless steel is widely used in various industries such as aerospace, chemical processing, and transportation [12]. Joining steels is widely used in industries like automobile, shipbuilding, construction, etc. due to the low cost and better mechanical properties. AISI 304 stainless steel has found extensive application in

automotive and aerospace, architectural, food process equipment, and kitchen appliances. It is the most versatile and widely used alloy of the stainless steel family. It has high corrosion resistance and ease of fabrication [16]. The Tungsten Inert Gas (TIG) or Gas Tungsten Arc Welding (GTAW) process is extensively used for joining thin sections of stainless steel. However, it is not useful in joining thick sections in a single pass. Activated TIG (A-TIG) significantly increases weld penetration up to 1.5 – 4 times in a single pass. Because of its deep penetration ability, A-TIG is the focus of research amongst researchers [12]. Well-established tungsten inert gas (TIG) welding process has the limitations of lesser penetration; however, process parameters (such as current and voltage) need to increase to attain deeper penetration, which, in turn, affects the material's property owing to increased heat energy. The use of activated fluxes generates extra heat energy with the same process parameters and causes deeper penetration and increased dilution [19]. To improve the TIG welding process's industrial efficiency and penetration ability, one method commonly applied is the use of activating flux with the welding process [1].

AISI 304 stainless steel, which is used in the aerospace industry, is a superior option to aluminum because of its tensile strength, toughness, shear modulus, and ability to withstand high temperatures in specific applications such as engine and exhaust components, landing gear components, and critical superstructure joints in aerospace structures. Due to its improved weld penetration and narrow weld bead ability, single-pass welding is the best method for joining 8mm thick or thicker AISI 304 steel plates. In recent years, aerospace manufacturers have been exploring alternatives to aluminum, and one of these alternatives is aerospace-grade stainless steel. AISI 304 steel is highly resistant to corrosion in aerospace applications due to its protective oxide layer, which prevents oxidation. Although stainless steel is heavier than aluminum, it has higher tensile strength, shear modulus, and melting point. In certain applications, aerospace-grade stainless steel may outperform aluminum. Some examples where AISI 304 generally outperforms aluminum are:

Engine and exhaust components: Aerospace components made of stainless steel can function better and have a longer useful life due to their resistance to corrosion and higher temperature tolerance.

Landing gear components: Stainless steel is significantly more suited to withstanding the impact pressures that landing gear equipment must withstand due to its high tensile strength and shear modulus.

Key superstructure joints: Aerospace stainless steel can be used to strengthen and lower the chance of failure during flight in critical joints located in the superstructure of an aircraft.

The oxygen content in activated flux has played a significant role in hot cracking and void formations in the weld [19]. The amount of ferrite in stainless steel weld metals can influence a wide range of properties, including corrosion resistance, toughness, long-term high-temperature stability, resistance to hot cracking, etc [6]. The certain range of oxygen content dissolved in the weld led to a significant increase in the penetration capability of TIG welds. TIG welding with active fluxes can increase the delta-ferrite content and improve the mechanical strength of the welded joint [2]. The vermicular ferrite in the ferrite-austenite (FA) mode in the weld microstructure is primarily dictated by the austenite phase's crystallographic directed interaction with ferrite, which is created at ferrite nucleation along the direction of heat flow. Oxide-based fluxes like SiO_2 , TiO_2 , and CrO_3 in pure argon shielding could act as insulators and that may form a barrier [2, 20]

in the path of current flow direction, therefore, the comprehensive study of their behavior is required to be studied. Argon shielding gas with oxide-based fluxes could enhance the penetration [20] and its effect on mechanical and microstructural properties in AISI 304 can be studied for further understanding, which can be employed by the aerospace industries for single pass full penetration welding.

The joints of AISI 304 steel that are anticipated to be characterized in this project are single pass, 8mm thick plate joints. It is anticipated that the joints will be fabricated under ideal circumstances using the A-TIG process, which is dependent on the mixtures of fluxes, flux solvents, and binders, as well as the ideal mixture of shielding gas environment and welding parameters.

• Definition of The Problem

The GTAW process is the most extensively used method for welding stainless steel because of its consistent heat source, superior weld quality, and affordable equipment [19]. The primary obstacles to using this technique for welding stainless steel in the aerospace, nuclear, and offshore industries—especially when the plate thickness is greater than 8 mm—are its shallow penetration, limited dilution, and low deposition rate in a single pass. According to GTAW, penetration is a crucial response characteristic that greatly regulates the autogenous welding production rate. Edge preparation is a necessary step before welding on pipe or plate thicker than 3 mm, and it takes a lot of time. The GTAW method is regarded as being uneconomical due to its lower penetration and high time consumption, even with excellent weld quality. In an effort to solve these issues, scientists have looked into the A-TIG technique, which uses activated flux to improve penetration without requiring edge preparation before welding [9, 11, 15, 19, 20]. Different materials work well with different kinds of active fluxes. In addition to flux, there are other welding parameters that affect the weld geometry and characteristics. To achieve the desired welding geometry, the right flux and optimal welding parameters must be used [13]. The combined action of reverse Marangoni convection and Lorentz forces [19, 20], which influence the behavior of ions and electrons during arcing [17, 18, 19], allows for deeper penetration in this process. Deeper penetration is predicted as a result of the arc constriction behavior mechanism [7, 11, 23] being promoted by the flux/compounds with enhanced oxygen contents. This changes the surface tension gradient and occasionally results in reverse Marangoni convection [8, 19, 20, 23]. When the weld oxygen concentrations approach 70×10^{-6} or higher, the surface tension-induced Marangoni convection shifts from an outward to an inward and large weld pool into a smaller and deeper one [9, 19]. The addition of CO_2 shielding [9] and the breakdown of oxide-based fluxes both contribute to the increased oxygen content in the weld [19, 20]. Together, the surface active components in the flux and the decomposed oxygen increase the heat produced in the weld, causing deeper penetration and a change in the shape of the weld pool due to the elevated temperature [4, 19, 20].

• Objectives

- To achieve a through-thickness single pass weldment of austenitic stainless steel AISI 304, fabricated using activated gas tungsten arc welding process with activated fluxes (SiO_2 , TiO_2 , and CrO_3) and sodium silicate (Na_2SiO_3) binder.
- To achieve the optimized welding parameters for the GTAW process for 8 mm thick rolled plates of austenitic stainless steel AISI 304 to obtain a successful through-thickness weld.
- To analyze the weld microstructure, weld surface macrographs, and the mechanical properties such as microhardness, tensile, and impact toughness properties of the activated GTAW weldment.

Chapter 2

Literature review

- **Introduction**

Brief reviews of the literature on research on TIG, A-TIG, and hybrid processes of various kinds on a range of stainless steels utilizing various welding settings and other factors are covered in this chapter to boost the productivity of the relevant process. In order to forecast and improve the mechanical characteristics, the welding community has conducted a great deal of study. In order to determine the impact of various process parameters on the mechanical characteristics and microstructures of various metals, researchers have experimented with TIG welding on a variety of steel and other metal types. Numerous researches have reported on the changes in weld penetration depth (weld morphology) and the causes that cause these changes. A few of the researchers studied the effects of various shielding gases and their mixes on various steel kinds. The available literatures are categorized in the following broad categories:

- Influence of Activated Flux on Microstructure and Mechanical Properties
- Influence of Current and Shielding Gas
- Influence of Process Parameters and Thermal Stress Analysis
- Influence of Hybrid Process

- **Literature Review**

- **Influence of Activated Flux on Microstructure and Mechanical Properties**

Tathgir, Rathod, and Batish [19] worked on the effect of activated fluxes in A-TIG welding of Duplex stainless steel and found that Oxide-based flux would govern the dendrite arm spacing and length of dendrites owing to the effect of heat produced and this can be influenced by the oxygen content in the weld. More oxygen content with higher heat generation dominates the higher grain size, long dendrite length, and higher dendrite arm spacing in the case of A-TIG welding with SiO_2 and CrO_3 flux.

Nakhaei, Khodabandeh, and Najafi [11] found that the presence of the active gas in the molten pool leads to the change in the temperature coefficient of surface tension from negative to positive value so that the Marangoni convection turns inward and forms a deep and narrow weld pool. Under these conditions, the maximum weld penetration of 2.5 mm is obtained for conventional TIG welding, while the deep weld profiles up to 6 mm are formed by the advanced A-TIG method.

Shyu et al. [15] found that the increases in weld depth and the decrease in bead width are significant with the use of the Cr_2O_3 , TiO_2 , and SiO_2 . The greatest improvement function in the penetration is with the

use of SiO_2 , but the Al_2O_3 led to the deterioration in the penetration and excessive slag compared with the conventional TIG process for stainless steel 304 welds. The CaO does not affect A-TIG penetration. **Baghel et al. [3]** found that V, U, and bowl-shaped weld beads were produced for welds without flux, oxide, and chloride flux, respectively. The highest aspect ratio was obtained with the use of oxide flux. Columnar dendrites, fine grains, and coarse grains were seen in the fusion zone, FGHAZ, and CGHAZ respectively whose morphology and size varied with welding conditions i.e. use of oxide and chloride flux and no flux during welding.

Vinothkumar et al. [22] found that the fractures of the tensile test sample occurred in stainless steel base metal which is evident that the strength of the welded joint was better than that of base metal. Tensile strength is good in the Al_2O_3 assisted welding. Microstructure tests show that A-TIG welding metal area with Al_2O_3 consists of grain growth and the CuO flux consists of voids. The presence of Al_2O_3 also produces a closely packed structure in the weld region.

Sharma and Dwivedi [14] worked on A-TIG welding of dissimilar P92 steel and 304H austenitic stainless steel and found that the microstructure of the weld joint possessed untempered lath martensite, prior austenite grain boundaries, ferrite, and polygonal austenite in various zones. The weld fusion zone possessed untempered lath martensite. The CGHAZ showed complete dissolution of precipitates and FGHAZ showed partial dissolution of precipitates. Coarse polygonal austenite grains and ferrite stringers were observed in 304H HAZ.

- **Influence of Current and Shielding Gas**

Tathgir, Rathod, and Batish [20] found that Dendrite arm spacing in ferrite is longer than austenite in unflux conditions and both pure Ar and H_2 -induced shielding conditions. However, there is a reduction in arm spacing of austenite and ferrite due to the presence of H_2 in shielding.

Tathgir, Bhattacharya, and Bera [18] found that a significant increase in depth of penetration and reduction in the width-to-penetration ratio is achieved by the ATIG process using a fine activated TiO_2 flux layer. Maximum of 37.8%, 44.3%, 47%, and 124% increase in depth of penetration for AISI 1020, AISI 304, AISI 316, and Duplex 2205 steels, respectively are measured while activated flux is used.

Tathgir and Bhattacharya [17] found that a significant increase in penetration occurs for different steels like AISI 304, AISI 316, Duplex 2205 and AISI 4340 steels when SiO_2 , MoO_3 , MoS_2 , CrO_3 and TiO_2 fluxes are used. A substantial increase in penetration depth is achieved using the activated fluxes that contain oxygen however the fluxes that contain chlorine, fluorine, or bromine like CaF_2 , NaF , KCl , KBr has insignificant effect and results in penetration even lower than no flux condition. The addition of H_2 in shielding gas aids higher heat conduction, enhances melting efficiency, and increases penetration.

Tusek and Suban [21] found that with hydrogen addition to the argon shielding gas the quantity of the austenitic stainless steel melted in TIG welding increases even by four times.

Durgutlu [5] found through penetration profile examinations for all three different shielding gases (pure argon, 1.5% H_2 -Ar, and 5% H_2 -Ar) show that penetration depth and weld bead width increases with increasing hydrogen content.

- **Influence of Process Parameters and Thermal Stress Analysis**

Pandya, Badgujar, and Ghetiya [12] found that an increase in welding current promotes the weld pene-

tration but an increase in welding speed and arc length results in a reduction of depth, width, and d/w ratio for both high and low sulfur steels. The cross-sectional area at weldment and electron temperature reaches a maximum value at an electron tip angle of 45° . A larger diameter of the electrode is more suitable due to the high electrode consumption rate in A-TIG welding.

Suraj et al. [16] found that welding current has more effect on depth of penetration, hardness, and width of weld bead than shielding gas flow rate and flux weight. The optimum value is obtained for maximizing the weld depth, and hardness and minimizing the weld bead width for welding current 200 Amp, shielding gas flow rate 13 L/min, and flux weight 0.149 gm by using response surface methodology optimization.

Modenesi, Apolinario, and Pereira [10] found that the use of a flux, even of extremely simple formulation, can greatly increase (up to around 300%) the weld penetration in TIG welding. It was possible to obtain full penetration welds in 5 mm thick plates of austenitic stainless steel with no preparation and currents of about 230 A. The operational characteristics of the ATIG process were not very different from those of conventional TIG welding. The arc voltage changed by $<3\%$, and, in most cases, the voltage variation was <1 V. Comparison of the operational parameters, of the arc shape, and the geometric parameters of the beads, suggests that factors acting on the weld pool are, possibly, responsible for the changes in weld bead shape.

• Influence of Hybrid Process

Lu, Fujii, and Nogi [9] found that Double shielded GTA welding using the pure inert Ar gas as the inner layer shielding and the active CO_2 or Ar- CO_2 gas as the outer layer shielding can protect well the tungsten electrode from oxidation in the welding process. The inner layer of Ar gas can effectively prevent the outer layer of active gas from contacting and oxidizing the tungsten electrode. The weld shape changes from a wide shallow one to a narrow deep one and the weld depth/width increases from 0.28 to 0.41 when the carbon dioxide is used as the out-layer shielding. Double-shielded GTA welding is a method to improve GTA welding production.

Kumar, Ahmad, and Singh [7] used a newly developed tri-component system (Cr_2O_3 , FeO , and MoO_3) as an activated flux for the efficient welding of Inconel 718 superalloy, and found that This unique combination

of flux increases the penetration depth almost twice as compared to C-TIG welding and seems quite better than the other previous work. It could be due to the presence of surface active metal oxide components of the flux system which changes the direction of Marangoni convection.

Xie et al. [24] worked on NSA-TIG welded AZ31 magnesium alloy joints and found that the increasing surface coating density of the NSA flux damaged the macro-appearance of the welding seam and increased the weld width and penetration due to the application of the TiO_2 . With the increase in the surface coating

density, the UTS of NSA-TIG welding joints increased firstly because of the refinement of α -Mg grains and distribution strengthening effects of particles including β - $\text{Mg}_{17}\text{Al}_{12}$, Mg_2Si and SiC , but then decreased rapidly due to the significant agglomerations of the nano-SiC particles.

• Summary of Literature

- The literature review presented in section 2.2 revealed that in the A-TIG process, the flux employed results in a molten metal inward flow due to the Marangoni effect. This flow is influenced by electromag-

netic forces, gravitational forces, and buoyancy forces, as well as the Lorentz force or drag forces caused by the movement of shielding gas over the weld pool. Additionally, the arc becomes more constricted, leading to deeper penetration.

- Reverse Marangoni convection is the name given to the phenomena of deep penetration caused by the arc's constriction and the shift in the surface tension gradient from negative to positive. The fluid flow of molten metal results from an outward direction to an inward direction.
- Authors discovered that using the face-bound approach increased penetration depth, however the penetration that was attained was narrower than with A-TIG welding. A few investigations have looked at the ideal flux amount for the face-bound method to maximize the penetration depth in austenitic stainless steel.
- In addition to tending to decrease angular distortion with increased mechanical strength, the A-TIG method aids in increasing penetration depth and the depth-to-width ratio.
- The impact of using H₂ in Ar as a shielding gas in MIG and TIG welding processes for various types of steel was examined by certain researchers. They discovered that the toughness value rises as the H₂ content in Ar increases, while the tensile strength increases up to a maximum of 1.5% addition of H₂ in Ar. H₂ in shielding gas immensely enhances DOP and more than 200-300% increase in penetration is noted as compared to pure Ar when TiO₂, SiO₂, or MoO₃, fluxes are used.
- At negative surface tension temperature coefficients below 2000K, the weld pool is broad and shallow; at positive values above 2000K, the weld pool is narrow and deep.
- The occurrence of Marangoni convection is demonstrated by authors to happen when the oxygen content in weld metal exceeds 70×10^{-6} .
- The oxidation behavior of the electrode during TIG welding with Ar and Ar+CO₂ double shielding was also studied by some authors. Their findings indicated that utilizing pure inert Ar gas for inner shielding and active CO₂ or Ar+CO₂ gas for outer layer shielding in double-shielded TIG welding effectively safeguards the tungsten electrode from oxidation, leading to improved penetration.

Chapter 3

Preliminary Investigation

3.1 Materials

AISI 304 stainless steel plates measuring 150 mm x 50 mm x 8 mm (length x width x thickness) were used in this welding experiment. The plates were cut to the required dimensions using a laser-cutting machine.

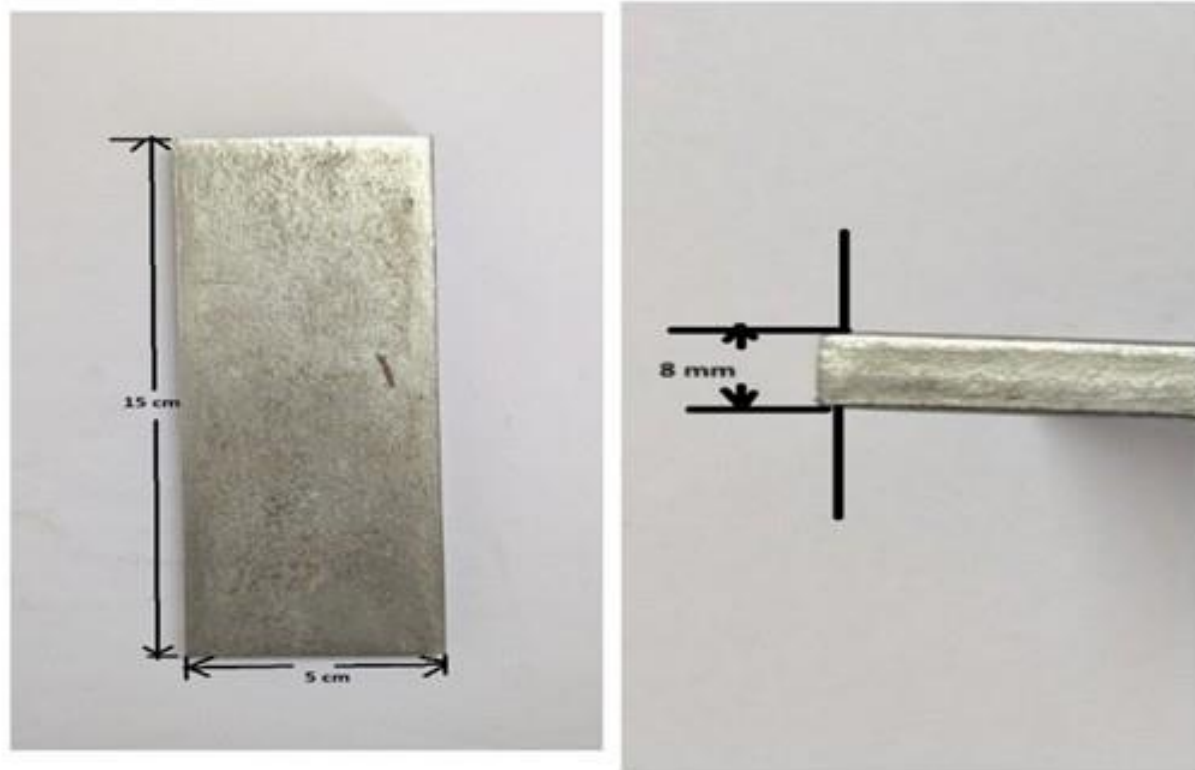
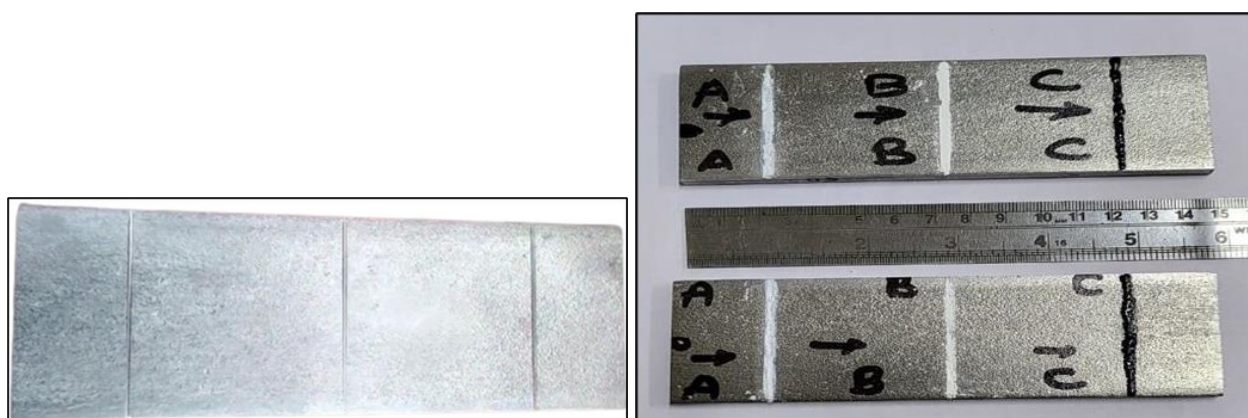


Figure 3.1: AISI 304 Plate After Laser Cutting

3.2 Plate Preparation for Preliminary Investigation

To ensure proper flux retention and minimize contamination, grooves were machined onto the plates using a shaper. The activated fluxes (SiO_2 (A), TiO_2 (B), and CrO_3 (C)) were then carefully pasted using Na_2SiO_3 binder onto the grooved surfaces and left to dry thoroughly. The plates were covered during drying to prevent dust or impurities from settling on the flux layer.



(a) Grooved Plate

(b) Flux Coated Plates

Figure 3.2: Plate Preparation for Preliminary Investigation

3.3 Experimental Setup

Initially, two welding speeds (50 mm/min and 60 mm/min) were considered for this experiment.

However, due to the limitations of the welding machine, only the higher speed of 60 mm/min was feasible. Therefore, the impact of varying current (150 A and 170 A) and utilizing individually activated fluxes on weld penetration and various weld properties was investigated while maintaining the constant welding speed. The welding process was executed with the following parameters, incorporating each activated flux separately in different

trials:

Preliminary Code	Weld	Current Condition	Activated Flux
A1		150 A	SiO ₂
B1		150 A	TiO ₂
C1		150 A	CrO ₃
A2		170 A	SiO ₂
B2		170 A	TiO ₂
C2		170 A	CrO ₃

Table 3.1: Preliminary Weld Coding



(a) Chalk Plastered Fixture with Clamping Nuts



(b) Robotic TIG Welding Machine

Figure 3.3: Weld Setup

3.4 Activated Flux Properties and Challenges

The activated fluxes were chosen for their distinct properties, but each presents unique challenges:

SiO₂(Silicon Dioxide) :

Properties: Improves fluidity, and reduces porosity.

Challenges: Excessive use can lead to slag inclusions and decreased weld strength.

TiO₂(Titanium Dioxide) :

Properties: It enhances arc stability, promotes deeper penetration, and improves toughness and crack resistance.

Challenges: It can be more expensive than other fluxes, and the potential for TiO₂ fume inhalation hazard (depending on the welding process and ventilation).

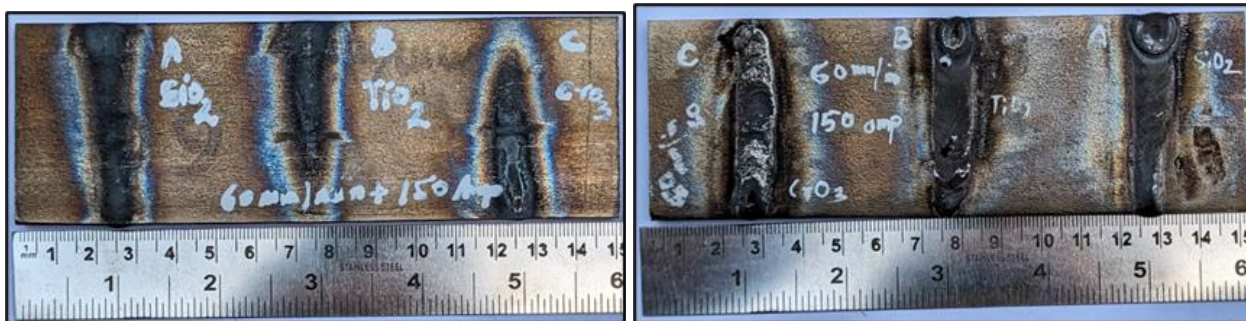
CrO₃(Chromium Trioxide) :

Properties: Powerful oxidizing agent, removes impurities, improves weld surface appearance.

Challenges: Toxic and requires careful handling due to hexavalent chromium content. Following the successful completion of bead-on-plate welding, a metallurgical analysis was conducted to investigate the weld penetration characteristics of three distinct fluxes: SiO₂, TiO₂, and CrO₃. The objective was to gain insight into the influence of these fluxes on weld profiles and penetration depths.

Welding Parameter/Condition	Value
Welding current [A]	150
Welding speed [mm/min]	60
Diameter of electrode [mm]	2.4
Stand-off distance [mm]	3
Shielding gas Argon Gas flow rate [L/mm]	10
Flux material	SiO ₂ (A), TiO ₂ (B), and CrO ₃ (C)

Table 3.2: Processing Condition (1) for Autogenous A-TIG Welding Experiment



(a) Front Side

(b) Back Side

Figure 3.4: Bead-on-Plate Welded at 150A and 60mm/min Welding Speed

Welding Parameter/Condition	Value
Welding current [A]	170
Welding speed [mm/min]	60
Diameter of electrode [mm]	2.4
Stand-off distance [mm]	3
Shielding gas Argon Gas flow rate [L/mm]	10
Flux material	SiO ₂ (A), TiO ₂ (B), and CrO ₃ (C)

Table 3.3: Processing Condition (2) for Autogenous A-TIG Welding Experiment



Figure 3.5: Bead-on-Plate Welded at 170A and 60mm/min Welding Speed

3.5 Methodology

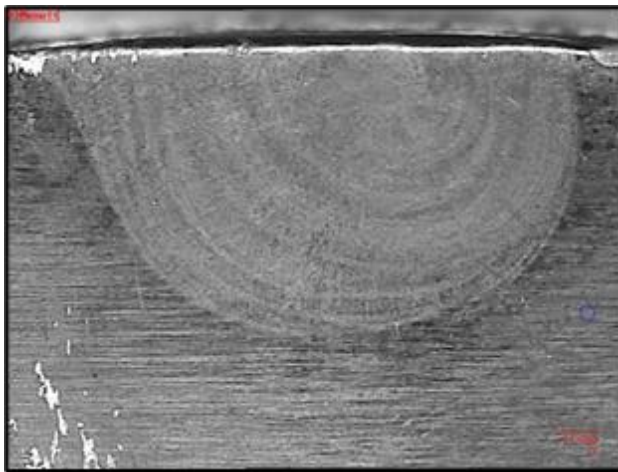
- Welded samples were sectioned transversely, ground using various Silicon Carbide grinding papers of different micron sizes, and polished using diamond paste abrasives to reveal the weld cross-section.
- Etching was performed using 14 Beraha's reagent which is a mixture of 20g Ammonium Bifluoride(NH₄HF₂) and 0.5-1g Potassium Metabisulphide(K₂S₂O₅) in 100ml of distilled water, to highlight microstructural features, including the fusion zone, HAZ, and base metal.
- Macrographs were examined under Digital Toolmakers microscope to assess weld penetration profiles and measure depths at various points along the weld centerline.
- High-resolution images were captured for analysis, with image software aiding precise penetration measurements.
- Microstructural analysis was done using optical microscopy. Clear microstructures were attained which explained the undergoing mechanisms in the preliminary welds.



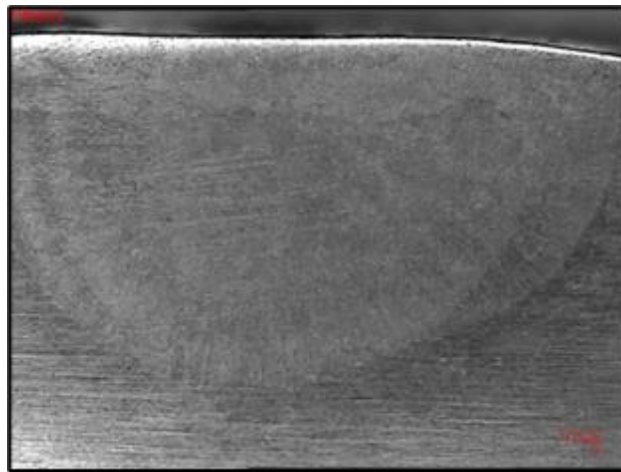
Figure 3.6: Preliminary Investigation Specimens

3.6 Results and Discussion

- **SiO₂ Flux** : This flux produced a weld with moderate penetration depth and a relatively wide fusion zone. SiO₂, being acidic, may have contributed to increased fluidity and wider spreading of the weldpool.
- **TiO₂ Flux** : Deeper penetration was observed compared to SiO₂, alongside a narrower fusion zone. TiO₂'s deoxidizing properties and lower viscosity likely facilitated improved penetration.



(a) SiO₂ at 150A



(b) SiO₂ at 170A

Figure 3.7: Preliminary Investigation Macrographs of SiO₂

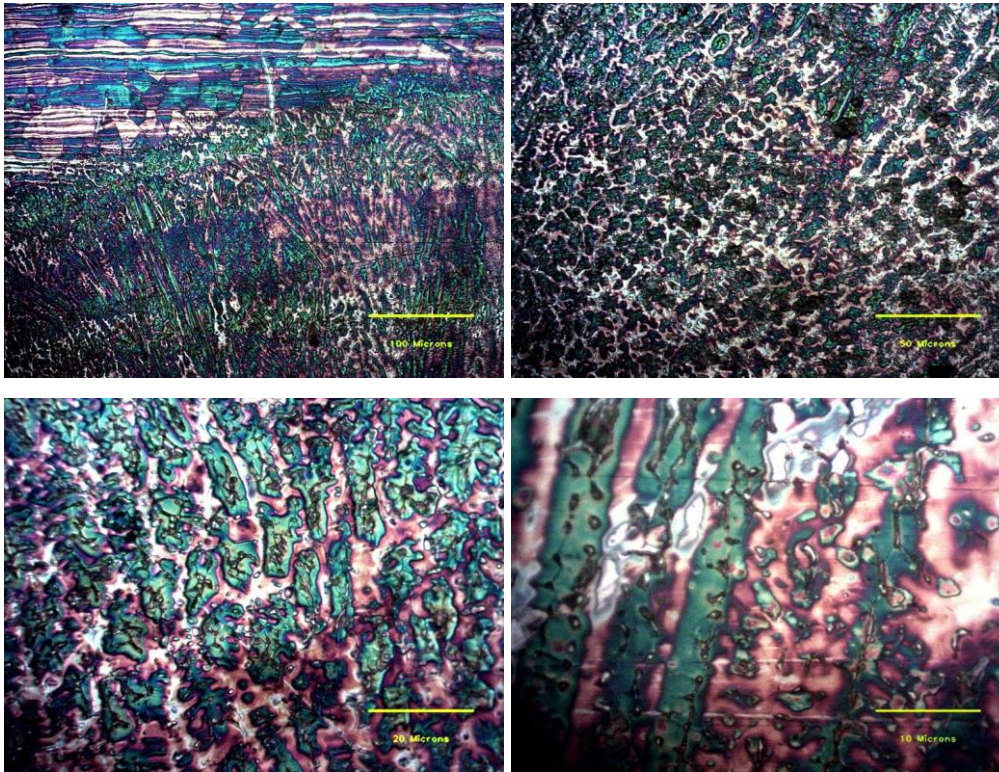


Figure 3.8: Preliminary Investigation Microstructures of SiO₂ at 150A weld

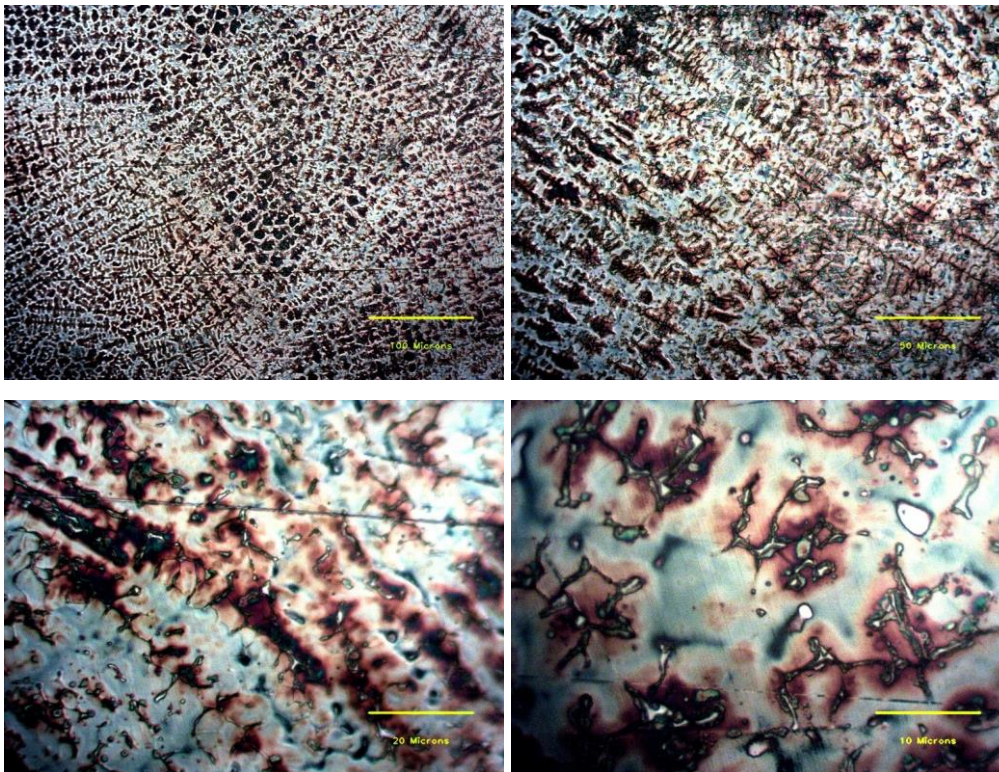
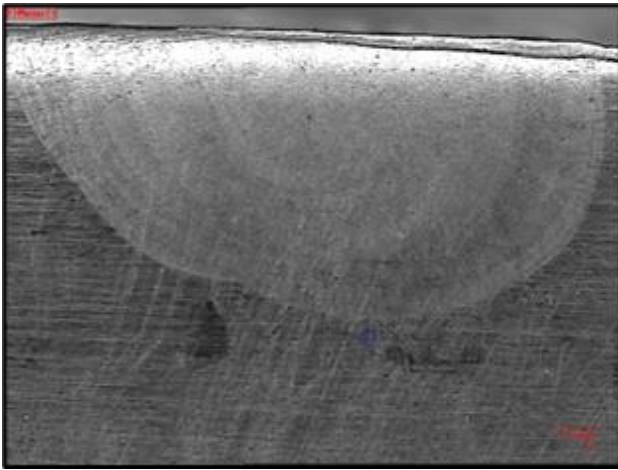
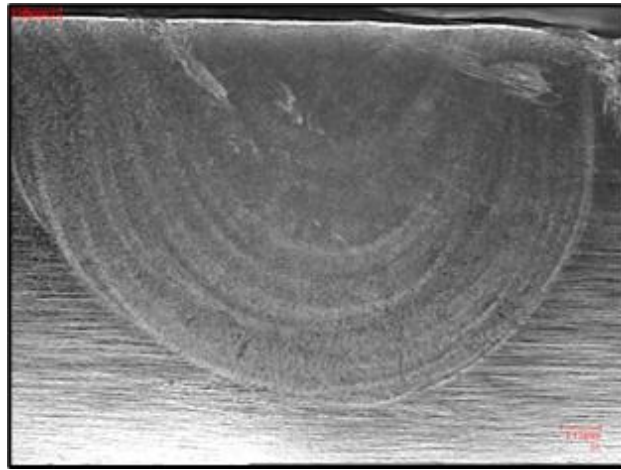


Figure 3.9: Preliminary Investigation Microstructures of SiO₂ at 170A weld



(a) TiO_2 at 150A



(b) TiO_2 at 170A

Figure 3.10: Preliminary Investigation Macrographs of TiO_2

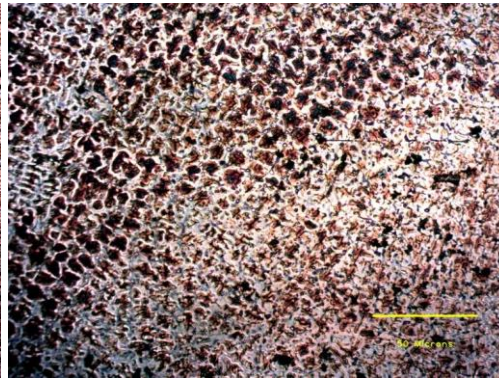
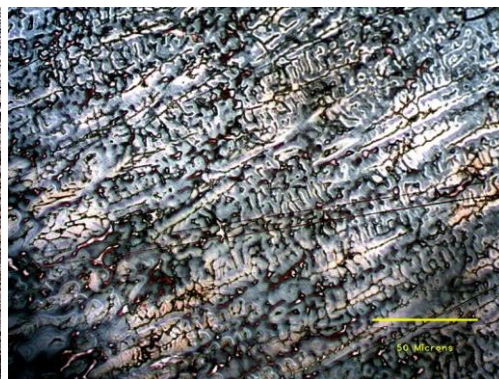


Figure 3.11: Preliminary Investigation Microstructures of TiO_2 at 150A weld



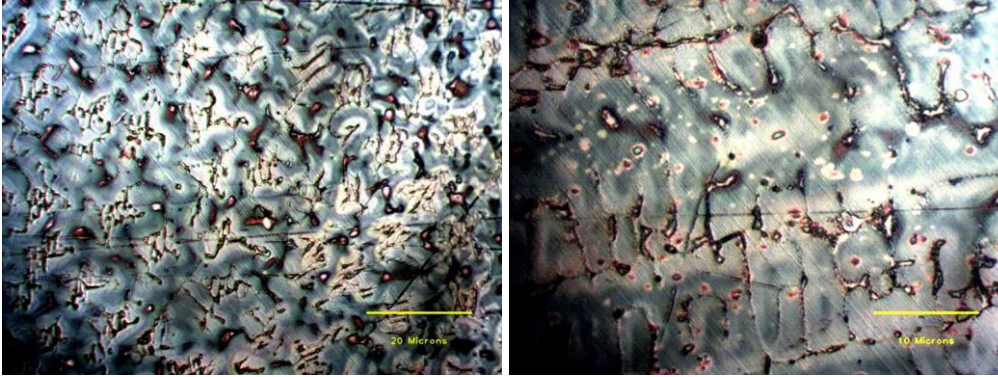
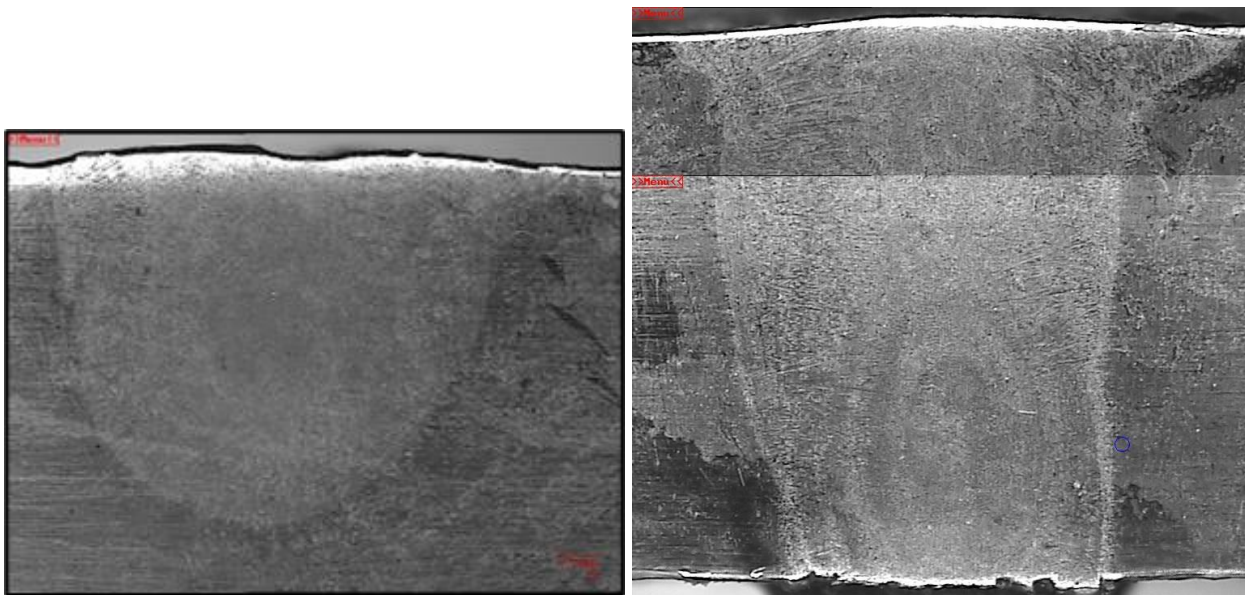


Figure 3.12: Preliminary Investigation Microstructures of TiO_2 at 170A weld

- **CrO_3 Flux :** The deepest penetration was achieved with CrO_3 , exhibiting a well-defined and narrow fusion zone. CrO_3 's strong oxidizing nature potentially led to focused arc energy and enhanced penetration.



(a) CrO_3 at 150A

(b) CrO_3 at 170A

Figure 3.13: Preliminary Investigation Macrographs of CrO_3

We can see the results, using fluxes (SiO_2 , TiO_2 , and CrO_3) we get penetration of 5.8 mm, 5.52 mm, 8 mm in SiO_2 , TiO_2 , and CrO_3 respectively at 150A, 60mm/min, and in case of 170A, 60mm/min parameters we get 7.9mm, 8mm, and 8mm penetration respectively.

The varying penetration depths are attributed to flux chemical compositions and physical properties. Alloying elements and deoxidizers influence weld pool fluidity, surface tension, and arc stability, all crucial factors in penetration behavior.

3.7 Conclusion from Preliminary Investigation

Macrographic analysis effectively evaluated weld penetration with SiO_2 , TiO_2 , and CrO_3 fluxes. Results confirmed a clear correlation between flux composition and penetration depth. This knowledge is valuable for flux selection in applications demanding controlled penetration. The preliminary investigation gave us

the insight to focus on 170 A welding current and CrO_3 flux because of promising results of a through penetration. Although TiO_2 and SiO_2 did not give us a through-penetration here, under 170 A conditions, they are still capable of deep penetrations. Hence, we anticipate all fluxes under 170 A and CrO_3 under 150A to give us the aspired results.

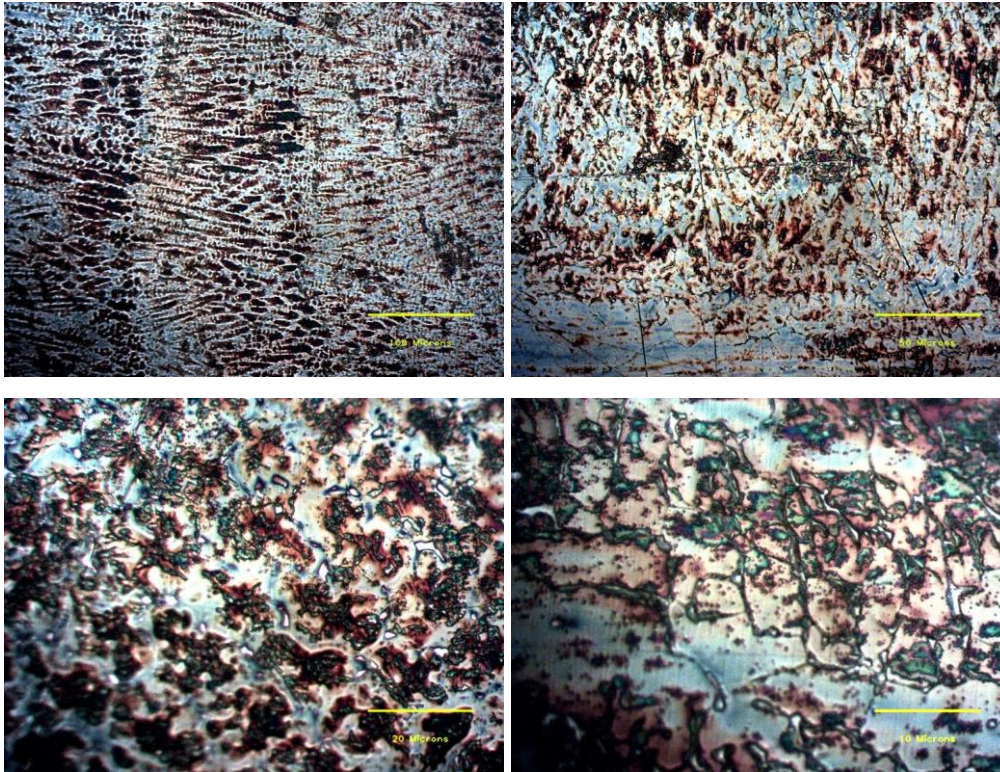
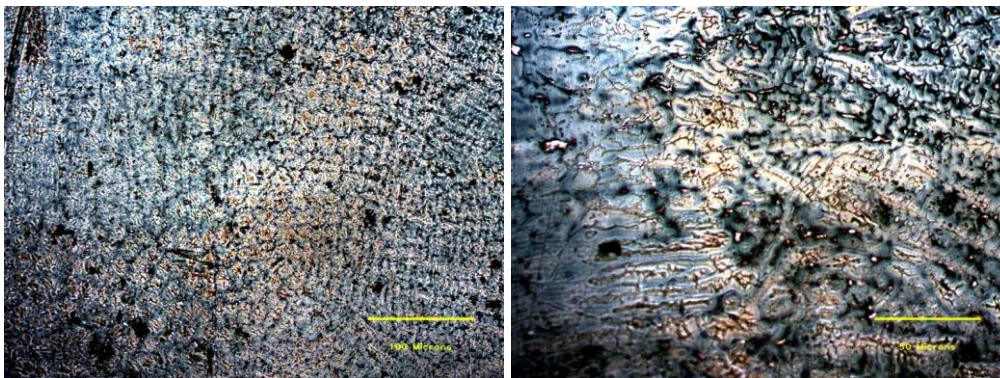


Figure 3.14: Preliminary Investigation Microstructures of CrO_3 at 150A weld



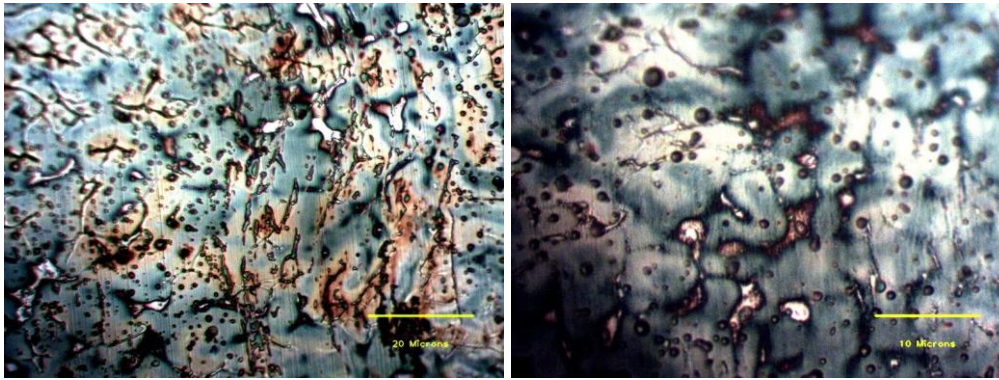


Figure 3.15: Preliminary Investigation Microstructures of CrO_3 at 170A weld The microstructural analysis provided us with insights on the HAZ and weld intersection. We can clearly see in the case of C1 and C2 (CrO_3 at 150, 170A) that there is an initialization of ferrite formation and that aligns with our knowledge of Cr being a Ferrite stabilizer. Also, we can see that as we approach the lower end of the weld, the density of ferrite formation decreases, which indicates that due to excessive heat generation of triple oxygen atoms in CrO_3 , Cr gets vaporized and does not penetrate deeply to form ferrite in deeper sections of the weld. Also, SiO_2 being an austenite favorable flux is also clear from the microstructures. Collected all the information, we may proceed on our project to study the activated flux TIG welding, we anticipate all our fluxes to give deep penetrations in 170A and CrO_3 flux to give us a through-thickness penetration.

